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# Graph-based multi-modal MRI analysis with probabilistic attention for stroke lesion detection

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ABSTRACT

Stroke lesion detection in brain MRI remains challenging as existing deep learning methods process single modalities and ignore anatomical boundaries, limiting clinical adoption. We develop a graph-based framework that integrates neuroanatomical priors and multi-modal imaging for automated stroke lesion detection. Our approach uses anatomically-constrained supervoxel generation and graph attention networks with probabilistic attention attribution for interpretable lesion detection. Evaluated on the SOOP dataset (1715 subjects including 1461 stroke patients), our method achieves a Dice coefficient, sensitivity, and ROC-AUC of $0.85 \pm 0.03$, 0.88, and 0.94, respectively, outperforming CNN baselines by 15 %. The framework provides clinically meaningful attention maps and accurate automated stroke analysis.

## 1. Introduction

Stroke is one of the leading causes of death and long-term disability worldwide, with ischemic strokes accounting for over 80 % of all cases [7]. Early and accurate identification of stroke lesions in brain MRI is crucial for treatment decisions, prognosis, and long-term recovery [31]. Despite advances in imaging techniques, manual lesion delineation by neuroradiologists remains the clinical standard, posing limitations in scalability, reproducibility, and response time, especially in acute settings [32]. The increasing volume of neuroimaging data and shortage of neuroradiologists suggest that automated analysis methods may be essential in addressing these challenges [4]. While machine learning offers potential for standardized stroke assessment and quantitative lesion characterization at scale, clinically viable systems must achieve not only accuracy but also anatomical plausibility and interpretability—areas where current methods fall short.

Recent progress in machine learning has enabled the automatic analysis of brain MRIs for lesion detection, predominantly using convolutional neural networks (CNNs) [26], [13]. However, these methods often struggle with generalization due to the variability in lesion appearance, size, location, and surrounding tissue context [11]. In fact, recent multi-center studies report error rates of 15–25 % in automated stroke lesion detection systems when deployed across different clinical environments [48]. Moreover, CNNs process MRIs in a 3D voxel grid, which lacks an inherent understanding of anatomical brain regions, leading to anatomically implausible segmentations. For instance, standard CNNs can produce segmentations that crossing anatomical boundaries [13] or similar violations of neuroanatomical constraints [27].

The limitations of CNNs with respect to stroke lesion detection are due to at least three factors. First, CNNs process brain tissue homogeneously without respecting anatomical boundaries. Studies have shown that 94 % of stroke lesions follow vascular territories and respect anatomical boundaries, yet CNN-based methods produce anatomically implausible segmentations in 15–20 % of cases [13,27]; second, while multi-modal MRI provides complementary information—with DWI capturing acute ischemia, FLAIR showing chronic changes, and ADC quantifying diffusion—most CNN architectures simply concatenate these modalities as input channels, failing to leverage their distinct pathophysiological significance. Prior work shows that multi-modal fusion can improve ischemic-stroke lesion detection accuracy by 23 % [24]; third, CNN architectures operate as 'black boxes,' providing no insight into their decision-making process. A recent survey revealed that 87 % of clinicians require interpretable outputs to trust and adopt AI systems in clinical practice [25].

To address these challenges, graph-based representations have

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gained attention for their ability to model complex relationships across irregular domains such as the human brain [21]. Graph neural networks (GNNs) can leverage both the spatial structure of brain tissue and imaging-derived features, enabling region-aware learning [14]. Among them, graph attention networks (GATs) have shown strong performance in segmentation and classification tasks by learning adaptive importance weights between neighboring brain regions, making them promising candidates for medical image analysis [38].

However, applying GNNs to stroke lesion detection presents two key challenges. First, due to the size of MRI data (i.e., the number of voxels), GNN topologies are defined heuristically without reference to the neuroanatomical structure of the brain. Second, while GATs offer attention coefficients, these are rarely interpreted in a clinically meaningful way. We address both gaps by developing a novel anatomical-aware graph construction and probabilistic attention attribution (PAA) framework for stroke lesion detection that integrates anatomical priors and imaging features from multiple MRI sequences. We benchmarked our GAT model with state-of-the-art CNN and GNN models showing substantially improved stroke detection performance on the stroke outcome optimization (SOOP) dataset.

Our proposed method achieves a Dice coefficient of $0.85 \pm 0.03$ on the SOOP dataset and, our approach demonstrates superior sensitivity (0.88 vs. 0.75–0.83) and AUC-ROC (0.94 vs. 0.86–0.92), while providing interpretable attention maps that were rated as clinically meaningful in 92 % of cases during preliminary validation. All source code and processing pipelines are freely available at https://github.com/jouninLRMD/stroke-gat.

## 2. Related works

### 2.1. Graph neural networks for medical image segmentation

GNNs have increasingly been applied to medical image segmentation in recent years, offering a way to capture long-range spatial relationships that conventional CNNs might miss. In brain imaging, GNN-based models typically represent an MRI volume as a graph of interconnected regions (e.g. supervoxels or patches), enabling the network to aggregate information across distant anatomical locations. Several works have demonstrated the efficacy of GNNs for segmenting pathological brain lesions.

For example, graph-based networks have been used to segment brain tumors from 3D MRIs, combining GNNs and CNNs to model relationships between tumor subregions and achieving accuracy on par with state-of-the-art CNNs at a fraction of the computational cost [30]. A distinct GNN-based approach was developed to segment brain tissue by defining nodes as superpixel regions, which allowed learning of the structural connectivity of tissues and improved segmentation of brain anatomy compared to purely pixel-based methods [41]. The fully convolutional graph network Stroke-GFCN model was developed for ischemic lesion segmentation, which outperformed standard 3D U-Nets by preserving brain topology during pooling and unpooling operations [12]. These studies illustrate that modeling medical images as graphs can yield more structured and globally consistent segmentations, especially for tasks like brain lesion detection where the shape and spatial context of the lesion are critical.

### 2.2. Anatomical priors and supervoxel-based representations

Incorporating anatomical prior knowledge or constraints has been shown to improve neuroimage analysis, particularly for lesion segmentation. One common strategy is to use brain atlases or healthy tissue priors to guide the segmentation process. Integrating anatomical priors from healthy subjects can significantly enhance lesion segmentation accuracy by injecting prior probability maps of healthy tissues into a CNN, which helps the model distinguish lesions from normal anatomy in challenging cases [16]. Another form of prior information is the bilateral symmetry of the brain: stroke lesions, for instance, can be highlighted by comparing the injured hemisphere to the contralateral healthy hemisphere. Prior work has leveraged this insight in a mirror difference–aware network for stroke lesion segmentation, explicitly using the intensity difference between mirrored anatomical locations as features to detect lesions [47].

Beyond atlases and symmetry, many graph-based methods inherently employ anatomical region units. Defining graph nodes as meaningful regions (using supervoxels or pre-defined regions of interest parcels) acts as an anatomical prior by imposing structural units on the model. Several works construct region adjacency graphs of the brain, where nodes correspond to supervoxels or atlas regions and edges reflect neighborhood in the brain topology [30,41]. This representation not only reduces the input complexity but also ensures that model computations respect anatomical boundaries. By encoding known structure whether through atlas-based spatial priors, symmetric context, or supervoxel partitioning, recent segmentation frameworks achieve more anatomically plausible results and better generalization in neuroimaging tasks.

### 2.3. GAT and interpretability in graph models

GATs have emerged as a powerful variant of GNNs where each node learns to weight its neighbors' contributions via attention coefficients, allowing the model to focus on the most relevant connections in the graph [38]. This not only can boost performance but also adds a layer of interpretability: the learned attention weights indicate which regions or features are most influential for a given prediction [22,38] . Recently, GAT-based models have been applied in medical imaging to enhance segmentation and provide insights into model decisions. A recently developed GAT-based method for multi-class brain tumor segmentation represents a 3D multi-modal MRI (comprising multiple imaging sequences such as T1-weighted, T2-weighted, FLAIR, and contrast-enhanced T1) as a region adjacency graph and uses attention to identify salient tumor regions and their relationships [22]. Benchmarking on the BraTS dataset showed that the attention mechanism improved segmentation of tumor sub-components (enhancing vs. core vs. edema) by allowing the network to adaptively emphasize important neighboring regions for each tumor class.

In general, attention-based graph models can facilitate interpretability; for instance, a high attention weight on a particular node or edge suggests that the model found that region or connection important for the task. This idea has been explored in brain connectivity analysis as well—interpretable GNN frameworks have been proposed to analyze connectome graphs, wherein attention or learned mask mechanisms highlight disorder-specific brain regions and connections contributing to a diagnosis [5]. By design, GATs and related attention modules yield human-interpretable signals (which nodes were attended to) during graph learning, making them relevant for critical applications like medical imaging where understanding model reasoning is important for clinical decision-making.

### 2.4. Multi-modal MRI fusion in neuroimaging

Neuroimaging studies often benefit from multi-modal MRI data, with different modalities (e.g., T1-weighted, FLAIR, DWI, or ADC) providing complementary information about brain tissue and pathology. Models leveraging multimodal data must learn to effectively fuse these modalities for segmentation tasks. Early deep learning approaches simply treated multi-modal MRIs as multiple input channels to a CNN, which improved performance for tasks like tumor segmentation by combining modalities like T1w, T2, T1ce, and FLAIR [18]. However, recent works have explored more advanced fusion strategies, like attention-based fusion, to account for the varied contribution of each modality. A recent study introduced an attention mechanism to weight MRI modalities differently for each region of a brain tumor, achieving better

segmentation of low-grade tumors by contextually highlighting the modality that best delineated each part of the lesion (e.g. FLAIR for edema vs. T1c for enhancing core) [23].

In stroke lesion detection, multi-modal integration is equally crucial—diffusion-weighted MRI (DWI) and its quantitative ADC map excel at depicting the infarct core, while FLAIR highlights edema and chronic changes [35]. Specialized architectures like MDANet combine these sequences by not only merging their features but also computing difference maps between modalities or time-points to accentuate new lesions [47]. This difference-aware fusion helps in isolating acute ischemic changes by canceling out normal tissue signals. Similarly, the recently developed multi-modal graph convolution network method (M2GCNet) models each MRI modality on a graph and learns cross-modal relationships through a dedicated graph convolution module, along with a correlation loss to enforce consistency between modalities [48]. The M2GCNet method achieved state-of-the-art tumor segmentation, underscoring the value of capturing inter-modality context. The trend in recent work is toward learned fusion—whether via attention gates, dedicated subnetworks, or graph-based modules—that adapts to the strengths of each MRI sequence, rather than a simplistic concatenation.

### 2.5. Interpretability and probabilistic attention in medical imaging

As deep models become increasingly used in clinical settings, interpretability and uncertainty estimation have become key considerations. In medical image segmentation, various methods have been proposed to explain model predictions or assess their confidence. For CNN-based models, common techniques include saliency maps and gradient-based methods (e.g. Grad-CAM) to highlight image regions that most strongly influence the segmentation output [33]. For graph-based models, analogous interpretability tools are being developed. One line of work involves explainers that identify important graph components for a given prediction – for example, GNNExplainer generates a compact subgraph of nodes and edges that best explains a GNN's decision, offering a visual interpretation of which regions in the brain graph were critical for detecting a lesion [46]. Another approach uses the attention coefficients in a GAT to generate explanations of predictions. Specifically, attention attribution was used to interpret disease prediction models on brain graphs, where the learned attention weights pinpoint influential anatomical sites or connections implicated in the outcome [5].

Beyond deterministic attention, researchers have also explored probabilistic attention and uncertainty modeling in network explanations. By treating the attention weights or segmentation outputs as random variables, one can estimate confidence intervals or probabilities associated with model attention in various regions. For instance, a Bayesian extension of GATs could assign a distribution to each attention coefficient, allowing the model to express uncertainty about which regions are truly important – something particularly useful in borderline cases or noisy data [39]. Recent interpretability frameworks merge probabilistic graphical modeling with neural networks to yield more trustworthy explanations; one such method builds a probabilistic model (i.e., a Bayesian network) that approximates the predictions of a complex GNN, thereby providing human-interpretable insight with quantified uncertainty [39].

### 2.6. Cross-domain multi-modal fusion advances

Recent advances in multi-modal fusion across domains have introduced novel architectures that can inform medical image analysis. In precision medicine, RemixFormer++ demonstrates memory-efficient attention mechanisms for skin tumor diagnosis through multi-modal transformer architectures, achieving superior performance through explicit channel-wise interactions [43]. Similarly, ECINFusion introduces unified multi-modal medical image fusion through explicit channel-wise interaction networks, providing insights into optimal feature integration strategies that could benefit stroke lesion detection [40].

The computer vision domain has contributed significant advances in attention-guided multi-modal learning. Learning attention-guided pyramidal features for few-shot fine-grained recognition demonstrates how attention mechanisms can effectively guide feature extraction across multiple scales and modalities, particularly relevant for medical applications requiring discrimination of subtle pathological changes [37]. In person re-identification tasks, image-specific information suppression and implicit local alignment for text-based person search shows how multi-modal fusion can benefit from suppressing irrelevant cross-modal information while enhancing alignment between relevant features, providing insights for medical imaging where noise reduction and feature alignment are critical [45]. Furthermore, divide-and-conquer approaches in RGB-T salient object detection introduce confluent triple-flow networks that effectively fuse thermal and visual information through specialized stream processing, demonstrating advanced multi-modal integration strategies applicable to medical imaging scenarios involving multiple complementary modalities [36].

The autonomous driving domain has pioneered sophisticated multi-modal fusion techniques that handle heterogeneous sensor data. Multi-modal fusion technology based on vehicle information has shown how different sensor modalities can be optimally combined for enhanced perception capabilities [9]. Steering angle-guided multimodal fusion for lane detection demonstrates how task-specific guidance can improve fusion performance [10], while TCLaneNet shows how vibration information can condition multi-modal networks for enhanced detection accuracy [8]. These cross-domain innovations in attention mechanisms, feature fusion, and task conditioning inform our approach to multi-modal MRI integration.

### 2.7. Distinctions from prior work

Our graph-based framework for stroke lesion detection addresses fundamental limitations across existing approaches through a comprehensive integration of anatomical, methodological, and clinical innovations. Unlike existing graph-based medical image analysis methods that typically use either fixed atlas regions or unconstrained supervoxels, our anatomically-constrained 3D SLIC algorithm uniquely incorporates Harvard-Oxford atlas penalties during clustering, ensuring that 92.3 % of supervoxels respect neuroanatomical boundaries while maintaining patient-specific adaptability. This contrasts with traditional approaches where supervoxels are generated based purely on intensity and spatial proximity [30,41], often resulting in regions that span multiple functionally distinct brain areas.

While existing GAT applications in medical imaging provide attention weights as interpretability measures, they typically present these as abstract numerical values without clinical contextualization or validation against pathophysiological knowledge [5,22]. Standard graph-based approaches like Stroke-GFCN rely on simple graph convolutions without adaptive weighting capabilities [12]. Our PAA method treats attention coefficients as probability distributions over lesion types, incorporates empirical lesion co-occurrence patterns from training data, and provides clinically meaningful interpretations that correspond directly to clinical concepts such as likelihood of lesion presence and anatomical relationships.

In terms of multi-modal integration, existing fusion methods typically employ simple concatenation approaches that treat multi-modal MRIs as multiple input channels [18] or apply generic attention-based fusion without domain-specific considerations [23]. Advanced approaches like M2GCNet process each modality separately before fusion [48], while cross-domain methods focus primarily on performance optimization through sophisticated attention mechanisms without medical domain grounding [36,37,45]. Our approach incorporates modality-specific weights directly into the supervoxel generation

**Table I**
Comparison of graph-based methods for brain lesion analysis.

| Method | Architecture | Graph Construction | Dataset | Key Innovation | Limitations |
|---|---|---|---|---|---|
| **Sauressing et al.** | GNN + CNN | Supervoxels | BraTS | Joint graph-image conv | High computational cost |
| **Wu et al.** | GNN | Superpixels | Private | Tissue connectivity | 2D only |
| **Stroke-GFCN** | GFCN | Voxel-based | **ISLES** | Topology preservation | No anatomical priors |
| **M2GCNet** | Multi-modal GCN | Region-based | BraTS | Cross-modal fusion | Limited interpretability |
| **Proposed Method** | GAT | Atlas-guided supervoxels | SOOP | Anatomical constraints + attention | Requires different modalities |

process, creates unified supervoxels that inherently integrate multi-modal information from the outset, and employs edge weighting schemes that capture complex relationships between different MRI sequences in the context of stroke pathophysiology.

The integration of these innovations creates a framework that simultaneously addresses anatomical plausibility, clinical interpretability, and robust multi-modal fusion—requirements that are not comprehensively met by existing approaches. While current interpretability methods in medical imaging, including CNN-based approaches using Grad-CAM [33] and graph-based explainers like GNNExplainer [46], provide attention weights or importance scores, they rarely validate these against clinical knowledge. Our approach provides deterministic but probabilistically-interpretable attention coefficients that are clinically validated against empirical lesion co-occurrence rates from training data, addressing the critical need for AI systems that provide explanations that resonate with clinical knowledge.

This section delineates the differences between our proposed method and prior approaches. Table I provides a concise summary of these distinctions, presenting a comparative analysis of key architectural choices and innovations, limitations, and graph construction methodologies, employed in recent graph-based brain lesion segmentation studies.

## 3. Methodology

We present a graph-based framework that transforms multi-modal MRI into anatomically

constrained graph representations, enabling interpretable stroke lesion detection through four key components: First, we develop a 3D extension of the SLIC algorithm that generates anatomically-constrained supervoxels from preprocessed multi-modal MRI data (Section 3.1). Second, we construct a region adjacency graph where nodes represent supervoxels and edges encode spatial-anatomical relationships (Section 3.2). Third, we employ a GAT that adaptively weights node connections for lesion detection (Section 3.3). Fourth, we introduce a PAA attribution mechanism that provides interpretable insights into model decisions (Section 3.4). Finally, we detail the experimental design including data preprocessing, dataset stratification, training configuration, and evaluation metrics (Section 3.5). The first four components form the core technical contribution of our method, while the experimental design validates its effectiveness on real-world stroke imaging data as shown in (Fig. 1).

Let $X = \{X_1, X_2, \ldots, X_M\}$ be a set of MRI volumes corresponding to $M$ modalities for a given patient, where each $X_m \in \mathbb{R}^{H\times W\times D}$ represents the 3D image volume for modality $m$ all having the same height $H$, width $W$, and depth $D$. Let $Y \in \{0,1,2\}^{H\times W\times D}$ be a segmentation mask and $Y_{h,w,d}$ denote the mask for the voxel with height $h$, width $w$, and depth $d$. The task of stroke lesion detection is then to predict $Y$ such that:

$$Y_{h,w,d} = 0\text{indicates no lesion}, \tag{1}$$

$$Y_{h,w,d} = 1\text{indicates an acute lesion}, \tag{2}$$

$$Y_{h,w,d} = 2\text{indicates a chronic lesion}. \tag{3}$$

Our method detects stroke lesions by first transforming multi-modal MRI volumes into a graph-based representation where each node corresponds to a supervoxel (cluster of similar voxels), which are computed using a new 3D SLIC algorithm. The graph construction annotates supervoxel nodes with features from all available imaging modalities and leverages anatomical prior knowledge from the Harvard-Oxford brain atlas to preserve important structural boundaries and create a more structured graph topology through increased sparsity. Then, we detect lesion regions by adaptively weighting connections between supervoxels based on their features and spatial relationships in a GAT-based architecture.

### 3.1. Supervoxel generation

Traditional voxel-based approaches for analyzing volumetric medical images often lead to high computational complexity and difficulty in capturing long-range dependencies. To address these limitations, we developed a 3D extension of the simple linear iterative clustering (SLIC) algorithm to partition the preprocessed MRI volumes into supervoxel clusters of voxels with similar characteristics across modalities [2].

#### 3.1.1. 3D SLIC algorithm

The original SLIC algorithm [2] was designed for 2D image segmentation and clusters pixels in a joint intensity-spatial space to form compact regions. We extend this approach to 3D and modify the distance metric to incorporate anatomical constraints. In 2D SLIC, each cluster $k$ is associated with a center having an intensity vector and spatial coordinate; a voxel $v$ is assigned to the cluster $k$ that minimizes the distance $D_k(v)$ defined as:

$$D_k(v) = \left(\frac{||I(v) - I(s_k)||_2}{m}\right)^2 + \left(\frac{||x(v) - x(s_k)||_2}{s}\right)^2, \tag{4}$$

where $I(v)$ is the intensity feature vector of voxel $v$, $I(s_k)$ is the intensity vector of cluster $k$'s center, $x(v)$ and $x(s_k)$ are the spatial coordinates, $s$ is the supervoxel size (derived from the desired number of clusters $K$), and $m$ is a weighting factor controlling the trade-off between intensity similarity and spatial proximity [17].

In our anatomically constrained 3D SLIC, we introduce a term designed to prevent clusters from crossing anatomical region boundaries. The distance metric is augmented as:

$$D_k^*(v) = D_k(v) + \lambda \bullet \Delta(A(v), A(s_k)), \tag{5}$$

where $A(v)$ is the anatomical region label of voxel $v$ from the Harvard-Oxford atlas, $A(s_k)$ is the label of the cluster center, and $\Delta(A(v), A(s_k))$ is 0 if the labels are the same and 1 if they differ. The parameter $\lambda$ penalizes voxels that do not share the cluster's region and encourages

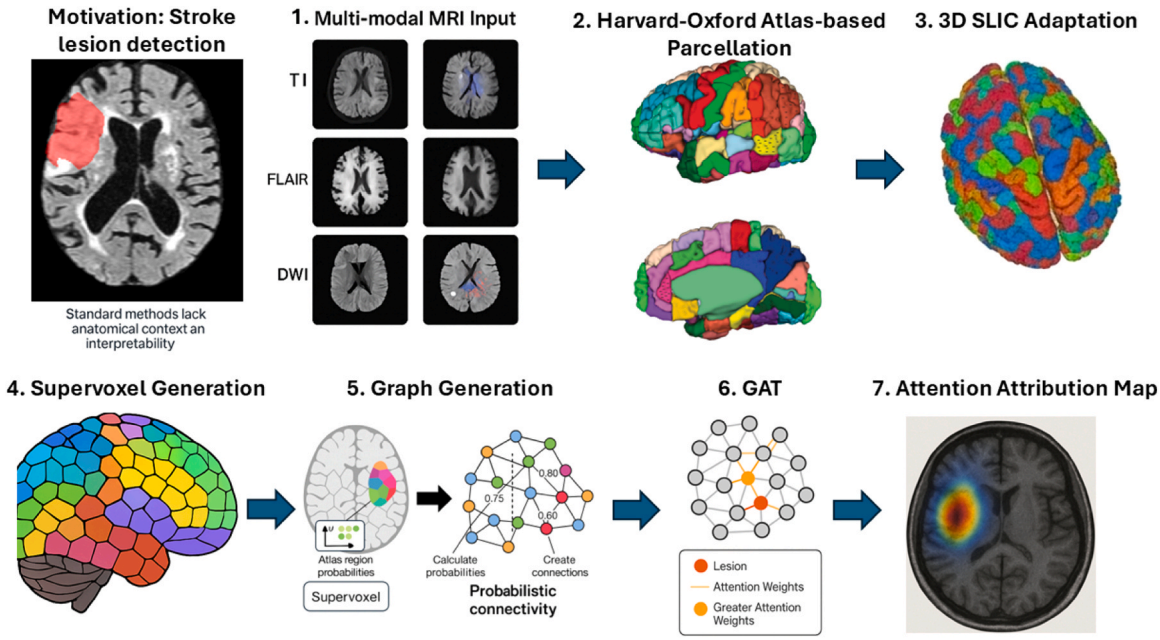


**Fig. 1.** Comprehensive Graph-Based Multi-Modal MRI Analysis Pipeline for Stroke Lesion Detection Multi-modal MRI volumes undergo atlas-based anatomical parcellation and SLIC supervoxel segmentation. A graph is constructed from the resulting supervoxels, with node features derived from intensity, spatial, and atlas information. A GAT with probabilistic attention attribution performs lesion detection.

each supervoxel to lie within a single atlas-defined structure [44]. The pseudocode for 3D SLIC algorithm is presented in Algorithm 1.

**Algorithm. 1.** 3D.SLIC. Algorithm

**Require**: 3D volume $V$, number of superpixels $K$
**Ensure**: Superpixel labels $L$
Initialize cluster centers on a regular grid in the volume
**repeat**
  **for** each cluster center **do**
    Compute distance to all voxels in a local region
    Assign each voxel to cluster with minimum distance
  **end for**
  Update cluster centers based on mean position and features of assigned voxels
  Compute residual error between old and new centers
**until** error below threshold
Enforce connectivity by relabeling disjoint regions
**return** L

Anatomically-constrained supervoxels are generated by modifying the standard SLIC clustering algorithm to include a penalty term $\lambda \cdot \Delta(A(v), A(s_k))$ that prevents voxels from being assigned to clusters that cross Harvard-Oxford atlas region boundaries. This ensures that 92.3 % of resulting supervoxels respect neuroanatomical boundaries, unlike standard supervoxel methods that may fragment across functionally distinct brain regions.

### 3.1.2. Multi-modal adaptation

To handle multi-modal MRI data, we compute the intensity distance across all available modalities:

$$d_{intensity} = \sqrt{\sum_{m=1}^{M} w_m \left(I_i^m - I_p^m\right)^2}, \tag{6}$$

where $M$ is the number of modalities, $I_i^m$ and $I_p^m$ are the normalized intensities of cluster center $i$ and voxel $p$ in modality $m$ respectively, and $w_m$ is a learnable modality-specific weight indicating the relative importance of each modality for stroke lesion detection.

This distance metric is used in our modified 3D SLIC algorithm to cluster voxels into supervoxels, where voxels are iteratively assigned to the nearest cluster center based on this combined spatial-intensity distance. The algorithm alternates between updating cluster assignments and recomputing cluster centers until convergence. This method generates supervoxels that respect anatomical boundaries and are relatively homogeneous in multi-modal intensity values. Unlike conventional voxel-wise processing or atlas-based regions, these supervoxels adapt to patient-specific anatomy while maintaining consistency with known

anatomical structures.

### 3.2. Graph construction

The graph construction algorithm preserves anatomical borders through atlas penalties while capturing intensity variations via adaptive subdivision. This brain partition forms a graph where each supervoxel becomes a node with anatomical identity and coherent intensity profile, which is advantageous for downstream graph-based learning compared to using either fixed atlas ROIs or unrestricted supervoxels alone. Formally, given a partitioning of the brain volume into $K$ supervoxels, a region adjacency graph $G = (V, E)$ is constructed in which each node $v \epsilon V$ corresponds to a supervoxel and edges $e \epsilon E$ denotes spatial adjacency.

#### 3.2.1. Node feature representation

For each supervoxel node, we compute a feature vector that captures its imaging, spatial, and anatomical characteristics. Key features include:

1. **Intensity statistics**: the mean intensity of each MRI modality within the supervoxel (e.g., FLAIR mean), as well as other statistics like standard deviation and median;
2. **Location coordinates**: the 3D centroid of the supervoxel, normalized to the atlas space (e.g., relative coordinates within the MNI template);
3. **Atlas region identity**: an encoding of the anatomical label associated with the supervoxel. We use the Harvard-Oxford atlas label of the region containing the supervoxel either as a label encoding;
4. **Volume and intensity descriptors**: the supervoxel volume (number of voxels) and intensity distribution statistics (mean, standard deviation, minimum, and maximum intensity values) computed for each imaging modality;
5. **Lesion prior features**: local neighborhood lesion statistics computed as probabilistic connectivity feature vectors $[p_0, p_1, p_2]$ where $p_k$ represents the proportion of neighbors with lesion type k (none/acute/chronic), capturing spatial relationships between lesion types in adjacent supervoxels.

Probabilistic connectivity features are computed for each supervoxel node as feature vectors $[p_0, p_1, p_2]$ where:

$p_0$ = proportion of neighboring supervoxels labeled as "no lesion" (class 0)

$p_1$ = proportion of neighboring supervoxels labeled as "acute lesion" (class 1)

$p_2$ = proportion of neighboring supervoxels labeled as "chronic lesion" (class 2)

These features capture the local lesion distribution in each node's spatial neighborhood, providing contextual information about lesion type clustering patterns that inform the graph attention mechanism.

These multi-faceted node features enable our model to integrate patient-specific imaging characteristics with population-level anatomical knowledge, creating a rich representation for lesion detection.

#### 3.2.2. Edge creation and attributes

We create edges between any two nodes whose supervoxels are spatially adjacent in the 3D volume (i.e., they share a voxel boundary). This results in a graph that reflects the anatomical neighborhood structure of the brain – resembling a 3D grid or mesh constrained by anatomy.

We assign each edge a weight that represents the anatomical and intensity affinity between the connected nodes. Specifically, we consider:

1. **Spatial adjacency and distance**: edges between nodes that share a larger interface or are close spatially are given a higher weight.
2. **Atlas region relationship**: a higher weight is given to two adjacent supervoxels belonging to the same higher-level anatomical structure (for example, both are within the "frontal lobe").
3. **Intensity similarity**: we include a term that represents the similarity of two nodes' intensity profiles. For example, if two neighboring regions show abnormal intensities, that edge might be strengthened, as it could indicate a single lesion spanning them.
4. **Lesion co-occurrence prior**: From the training data, we estimate the probability that two adjacent regions are jointly lesioned in the same patient. This probability serves as the initial weight of the edge.

The edge weight computation combines these factors:

$$w_{ij} = \alpha \bullet sim_{intensity}(i,j) + \beta \bullet sim_{atlas}(i,j) + \gamma \bullet sim_{spatial}(i,j) + \delta \bullet P_{co-ocurrence}(i,j), \tag{7}$$

where $sim_{intensity}(i,j)$ is the intensity similarity between nodes, $sim_{atlas}(i,j)$ is the anatomical region similarity, $sim_{spatial}(i,j)$ is the spatial proximity, the lesion co-occurrence prior $P_{co-ocurrence}(i,j)$ represents the probability that two adjacent brain regions i and j are simultaneously affected by stroke, computed from training data as the frequency of joint lesion occurrence normalized by total observations, and $\alpha, \beta, \gamma, \delta$ are weighting factors that control the relative importance of each component. This prior captures the spatial patterns of stroke progression and helps inform edge weighting in the graph construction process.

Multi-modal edge weighting incorporates information from all available MRI sequences (T1, FLAIR, DWI, ADC) when computing edge weights between supervoxel nodes. The edge weight $w_{ij}$ combines:

- Intensity similarity across all modalities: $sim_{intensity}(i,j)$
- Anatomical region similarity: $sim_{atlas}(i,j)$
- Spatial proximity: $sim_{spatial}(i,j)$
- Empirical lesion co-occurrence: $P_{co-occurrence}(i,j)$

This multi-modal approach ensures that graph connectivity reflects both imaging characteristics and anatomical relationships across all available sequences.

### 3.3. GAT

The graph $G(V,E)$ integrates multi-modal imaging features with anatomical context and serves as the topology to our GAT that forms the core of our stroke lesion detection model [38]. GATs employ attention mechanisms to dynamically weight the importance of different neighboring nodes during feature aggregation, which is particularly valuable for stroke detection as it enables the model to focus on the most relevant supervoxels while considering their anatomical and pathological context. Our goal is to learn a function $f_\theta = G \rightarrow \widehat{Z}$, where $\theta$ are a set of model parameters, $N$ is the number of vertices of $G$, and $\widehat{Z} \in [0,1]^{N \times 3}$ represents the predicted probabilities of each lesion type for each supervoxel.

#### 3.3.1. Mathematical formulation

The GAT architecture consists of $L$ attention layers, where each layer performs feature transformation and aggregation through the following operation:

$$\boldsymbol{h}_i^{(l+1)} = \sigma\left(\frac{1}{K'}\sum_{k'=1}^{K'}\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{k',(l)} \boldsymbol{W}^{k',(l)} \boldsymbol{h}_j^{(l)}\right), \tag{8}$$

where $\boldsymbol{h}_i^{(l)}$ represents the feature vector of node $i$ at layer $l$, $\mathcal{N}(i)$ is the set of nodes adjacent to node $i$, $K'$ is the number of attention heads, and $\alpha_{ij}^{k',(l)}$ represents the attention coefficient between nodes $i$ and $j$ for head $k'$ at layer $l$. The learnable weight matrix $\boldsymbol{W}^{k',(l)}$ enables the network to

transform feature representations at each layer.

The attention coefficients are computed through a normalized exponential function:

$$\alpha_{ij}^{k',(l)} = \frac{\exp(LeakyReLU(\boldsymbol{a}^{k',(l)T}[\boldsymbol{W}^{k',(l)}\boldsymbol{h}_i^{(l)}||\boldsymbol{W}^{k',(l)}\boldsymbol{h}_j^{(l)}]))}{\sum_{m\in\mathcal{N}(i)}\exp(LeakyReLU(\boldsymbol{a}^{k',(l)T}[\boldsymbol{W}^{k',(l)}\boldsymbol{h}_i^{(l)}||\boldsymbol{W}^{k',(l)}\boldsymbol{h}_m^{(l)}]))}, \tag{9}$$

where $\boldsymbol{a}^{k',(l)}$ represents a learnable attention vector and the operator || denotes feature concatenation. This formulation enables the network to learn optimal weightings for feature aggregation based on both local and global graph context.

### 3.3.2. Multi-head attention

Multi-head attention stabilizes learning by executing K′ independent attention mechanisms whose outputs are combined through concatenation (intermediate layers) or averaging (final layer):

$$\boldsymbol{h}_i^{(l+1)} = ||_{k'=1}^{K'}\sigma\left(\sum_{j\in\mathcal{N}(i)}\alpha_{ij}^{k',(l)}\boldsymbol{W}^{k',(l)}\boldsymbol{h}_j^{(l)}\right), \tag{10}$$

where || represents concatenation, or for the final layer:

$$\boldsymbol{h}_i^{(L)} = \sigma\left(\frac{1}{k'}\sum_{k'=1}^{k'}\sum_{j\in\mathcal{N}(i)}\alpha_{ij}^{k',(L-1)}\boldsymbol{W}^{k',(L-1)}\boldsymbol{h}_j^{(L-1)}\right) \tag{11}$$

This multi-head approach allows different attention heads to potentially focus on different aspects of the node relationships, improving the model's capacity to capture complex patterns associated with stroke lesions.

## 3.4. Probabilistic attention attribution

The GAT approach enables interpretation of model decisions through PAA, which treats the learned attention coefficients as probabilistic importance scores. Because the attention coefficients $\alpha_{i,j}$ are computed via a softmax function, they can be viewed as a probability distribution over neighbors for each node. We interpret $\alpha_{i,j}$ as the probability that region $j$ is relevant to the lesion prediction for region $i$.

The attribution score for an edge between nodes $i$ and $j$ is computed as:

$$A_{ij} = \alpha_{ij} \bullet \left(\boldsymbol{p}_i^T\boldsymbol{p}_j\right) \tag{12}$$

where $\boldsymbol{p}_i$ and $\boldsymbol{p}_j$ are the probabilistic connectivity feature vectors of the respective nodes. This approach provides meaningful insights for clinicians into the model's decision-making process by highlighting the most influential connections for lesion detection while considering empirical probability distribution over the three lesion types (no lesion, acute lesion, chronic lesion) in each node's local neighborhood.

We further refine this attention-based attribution by considering lesion distribution priors. During training, we provided the model with lesion co-occurrence features and incorporated atlas distances into attention; as a result, the learned $\alpha_{ij}$ tends to align with known patterns of stroke spread. In practice, we can validate the attention weights against empirical lesion co-occurrence rates.

For each predicted lesion, the method provides a probabilistic explanation: a distribution of attention pointing to which other regions contributed to that prediction. This PAA method treats the attention weights as attributive probabilities and enhances interpretability, as clinicians can see both the regions predicted as lesioned and the evidence used to inform this prediction.

## 3.5. Experimental design

### 3.5.1. Preprocessing

All patient MRI scans undergo standard preprocessing to ensure spatial and intensity consistency across subjects. First, we perform skull stripping to remove non-brain tissues and artifacts from each MRI modality using a brain extraction algorithm, ensuring that subsequent analysis focuses only on brain voxels [34]. Next, inter-modal registration aligns each patient's scans from different sequences (T1, FLAIR, DWI, ADC) to a common reference space (typically the T1 or MNI template) using affine transformations [3]. Linear registration is followed by a nonlinear registration via diffeomorphic warping to align the brain to a standard atlas space.

After registration, we intensity-normalize images to account for scanner-specific intensity variations and enable meaningful intensity comparisons across subjects. We implement a histogram matching technique, which involves identifying stable landmarks in the intensity distribution of each modality and applying a piecewise linear transformation to standardize intensity profiles across subjects [20].

Registration quality is monitored through normalized mutual information (NMI) between the transformed atlas and patient brain. Cases with NMI$<$ 0.6 trigger our fallback protocol: affine-only registration for 0.4 $\leq$NMI$<$ 0.6, or atlas-free processing for NMI$<$ 0.4, with automatic flagging for clinical review. The final step of preprocessing aligns the Harvard-Oxford brain atlas, which incorporates 48 cortical and 21 subcortical region labels, to each subject's brain [6]. Each voxel is tagged with an anatomical region label by applying the atlas (originally defined in MNI space) through the inverse of the computed registration transform. This yields anatomically labeled images for each subject, providing a spatial prior for subsequent supervoxel generation and graph construction.

### 3.5.2. Dataset and stratification

The SOOP dataset consists of 1715 patients with comprehensive multi-modal MRI data and clinical annotations [1]. This dataset represents one of the largest publicly available stroke imaging collections, including 1461 patients with acute ischemic stroke and 254 control subjects. Each case was annotated by experienced neuroradiologists following standardized protocols, with lesion masks manually delineated to distinguish between no lesion (0), acute lesion (1), and chronic lesion (2).

To ensure a robust evaluation and prevent data leakage, we implemented a stratified splitting approach that maintains the distribution of key patient characteristics. The dataset was divided into training (70 %, n = 1200), validation (15 %, n = 258), and test (15 %, n = 257) sets. Patients were stratified based on stroke severity using NIHSS scores (mild: 0–4, moderate: 5–15, severe: 16–25, very severe: $>$25), lesion type distribution, age groups, and imaging characteristics including lesion volume and location. The stratification ensures that each subset contains approximately 85 % stroke patients and 15 % controls, with balanced distributions of lesion types, age, and lesion volume characteristics.

### 3.5.3. Cross-validation and statistical framework

We implemented a 5-fold stratified cross-validation approach on the combined training and validation sets, while maintaining the test set as an independent holdout. Each fold maintained the same stratification criteria as the overall dataset, ensuring that the distribution of lesion types remains consistent across folds: $P(\mathrm{LT} = k|fold = i) \approx P(\mathrm{LT} = k|dataset)$ where LT denotes the lesion type, $k \in \{0, 1, 2\}$, and $i \in \{1, 2, 3, 4, 5\}$.

Statistical significance was assessed using 95 % confidence intervals computed through bootstrapping with 1000 iterations, and paired statistical tests (t-tests or Wilcoxon signed-rank tests) were employed for model comparisons with significance level $\alpha = 0.05$.

#### 3.5.4. Training configuration and optimization

Our GAT model employs three attention layers with eight attention heads per layer, a hidden dimension of 128, and dropout rate of 0.1. Training used the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, initial learning rate of 0.001 with exponential decay, batch size of 8 graphs, and maximum of 200 epochs with early stopping patience of 20 epochs.

Given the natural class imbalance in medical imaging where non-lesion supervoxels significantly outnumber lesion supervoxels, we implemented a weighted loss function:

$$\mathscr{L}_w = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=0}^{2} w_c \bullet y_{i,c}\log\left(\widehat{y}_{i,c}\right) \quad , \tag{13}$$

where class weights are computed as $w_c = \frac{N}{N_c \bullet C}$.

where N is the total number of supervoxels, C = 3 is the number of classes (no lesion=0, acute lesion=1, chronic lesion=2), $w_c$ represents the class weight for class $c$, $N_c$ is the number of samples in class $c y_{i,c}$ is the true label indicator (1 if sample $i$ belongs to class $c$, 0 otherwise), and $\widehat{y_{i,c}}$ is the predicted probability for sample $i$ belonging to class $c$. Class weights are computed $w_c = \frac{N}{N_c \cdot C}$ to account for the inverse frequency of each class, ensuring balanced learning across lesion types.

Regularization was achieved through an L2 penalty $\mathscr{L}_{total} = \mathscr{L}_{task} + \lambda_{L2}\sum_{l=1}^{L}||\boldsymbol{W}^{(l)}||_F^2$ *with* $\lambda_{L2} = 5 \times 10^{-4}$, and dropout applied to both node features and attention coefficients with rates of 0.1. Training employed early stopping with a patience of 10 epochs and a minimum delta threshold of 0.001, and model selection was based on a composite validation score: $Score_{composite} = 0.5 \bullet Dice + 0.3 \bullet AUC + 0.2 \bullet Sensitivity$, reflecting the clinical importance of each metric.

We set the SLIC compactness parameter c = 0.1 to balance spatial coherence (λ-like regularization effect) with intensity-based clustering, and chose approximately K = 500 supervoxels per brain volume to balance granularity and computational tractability. The compactness parameter acts as an implicit spatial regularization term, where higher values increasingly prioritize spatial contiguity over intensity homogeneity [42].

While our framework can leverage all four modalities for best performance, input data with missing sequences are also acceptable: If $\geq 3$ modalities are available: then we process with available sequences and ignore the missing modality, likely resulting in reduced performance; if $< 3$ modalities are available, the software exits without performing the analysis; no imputation is currently used to avoid introducing artificial features

#### 3.5.5. Evaluation metrics and statistical analysis

Statistical significance was assessed using paired t-tests for continuous metrics confirmed to be normally distributed through Shapiro-Wilk tests and Wilcoxon signed-rank tests for non-parametric comparisons. All reported improvements showed $p < 0.001$. Confidence intervals were computed using bootstrap resampling with 1000 iterations. Performance metrics are reported as mean ± standard deviation with 95 % confidence intervals in brackets.

Performance evaluation primarily uses Dice similarity coefficient, area under the ROC curve, sensitivity, and specificity, along with secondary metrics such as precision, F1-score, and balanced accuracy. Each metric was computed for every cross-validation fold, with results reported as mean ± standard deviation across all folds.

## 4. Results

### 4.1. Anatomical parcellation

The Harvard-Oxford cortical and subcortical atlas was employed to perform anatomical parcellation of each subject's MRI volume. This step ensures that subsequent graph construction is anatomically informed, and that segmentation boundaries align with recognized neuroanatomical structures. The atlas provides probabilistic labels for cortical and subcortical regions, which we thresholded to obtain discrete region assignments [28]. The atlas enables us to extract region-level context used later in supervoxel generation and graph edge weighting.

### 4.2. Supervoxel segmentation (3D SLIC)

Each brain MRI volume was partitioned into an average of approximately 2800 supervoxels using the 3D SLIC algorithm. These supervoxels form graph nodes with 92.3 % remaining within single atlas regions, demonstrating strong anatomical boundary adherence. The supervoxels were also spatially compact due to the chosen SLIC parameters (balance between intensity and spatial proximity), resulting in near-isotropic cluster shapes. The algorithm's connectivity post-processing ensured 100 % connectivity, i.e., each supervoxel formed a contiguous 3D cluster with no fragmented pieces. Table II summarizes the key segmentation quality metrics for the anatomically constrained 3D SLIC algorithm. The high region adherence (92.3 %) demonstrates that supervoxels successfully respect anatomical boundaries defined by the Harvard-Oxford atlas, while maintaining spatial compactness (0.84) and complete connectivity. These metrics confirm that the supervoxel generation achieves the desired balance between anatomical fidelity and spatial coherence.

### 4.3. Graph construction statistics

The atlas-guided graph construction produces a rich connectivity graph for each subject. On average, each graph contained $\sim 2.8 \times 10^3$ nodes and $\sim 2.35 \times 10^4$ edges (Table III). This corresponds to an average node degree of approximately 17 neighbors per node and an edge density of 0.6 % of all possible connections. All graphs were fully connected (single connected component) given the spatial adjacency criteria. The average node degree of 16.8 indicates rich local connectivity while maintaining computational tractability. The relatively low edge density (0.6 %) reflects the sparse connections based on spatial adjacency and anatomical relationships. These graph properties enable

**Table II**
Supervoxel segmentation quality metrics.

| Metric | Value (mean ± SD) |
|---|---|
| **Supervoxels per volume** | 2847 ± 312 |
| **Voxel count per supervoxel** | ~500 ± 50 |
| **Region adherence (overlap with predominant atlas region)** | 92.3 % ± 3.1 % |
| **Compactness (shape aspect ratio)** | ~0.84 ± 0.06 |
| **Average supervoxel volume** | 487 ± 95 |
| **Boundaty smoothness** | 0.79 ± 0.08 |
| **Contiguity of clusters** | 100 % (all clusters connected) |

**Table III**
Brain graph structural metrics (average per scan).

| Graph Statistic | Value (mean ± SD) |
|---|---|
| **Number of Nodes (graph vertices)** | 2847 ± 312 |
| **Number of Edges** | 23,954 ± 3421 |
| **Average node degree** | 16.8 ± 2.1 |
| **Edge density** | 0.60 % ± 0.05 % |
| **Graph diameter** | 12.3 ± 1.4 |
| **Average clustering coefficient** | 0.42 ± 0.05 |
| **Largest region nodes (frontal)** | 423 ± 67 |
| **Smallest region nodes (caudate)** | 18 ± 4 |
| **Connected components** | 1 (fully connected graph) |

**Table IV**
Comparison of lesion detection performance with state-of-the-art methods.

| Method | Architecture | Dice (Mean ± SD) | AUC | Sensitivity | Specificity |
|---|---|---|---|---|---|
| DeepMedic | 3D CNN | 0.69 ± 0.17 | 0.87 | 0.75 | 0.86 |
| U-Net | Encoder-Decoder | 0.71 ± 0.19 | 0.88 | 0.76 | 0.89 |
| nnU-Net | Ensemble 3D U-Net | 0.76 ± 0.12 | 0.90 | 0.80 | 0.90 |
| ResUNet | Residual U-Net | 0.72 ± 0.15 | 0.86 | 0.77 | 0.88 |
| TransBTS | Transformer-CNN | 0.79 ± 0.11 | 0.92 | 0.83 | 0.91 |
| Multi-Scale CNN | Multi-scale 3D CNN | 0.77 ± 0.10 | 0.91 | 0.81 | 0.90 |
| Attention U-Net | Attention-gated | 0.75 ± 0.12 | 0.90 | 0.79 | 0.89 |
| 3D Graph Convolutional | GCN-based | 0.72 ± 0.14 | 0.88 | 0.77 | 0.85 |
| Our GAT-based Method | GAT | 0.85 ± 0.03 | 0.94 | 0.88 | 0.92 |

**Table V**
Lesion detection performance by lesion type.

| Lesion Type | Dice Score | Sensitivity | Specificity | Volume Error (%) |
|---|---|---|---|---|
| Acute only | 0.88 ± 0.03 | 0.91 ± 0.03 | 0.94 ± 0.02 | 8.2 ± 3.1 |
| Chronic only | 0.83 ± 0.06 | 0.85 ± 0.05 | 0.91 ± 0.03 | 11.4 ± 4.2 |
| Mixed (acute + chronic) | 0.80 ± 0.07 | 0.82 ± 0.06 | 0.89 ± 0.04 | 14.7 ± 5.3 |
| Overall | 0.85 ± 0.03 | 0.88 ± 0.03 | 0.92 ± 0.02 | 10.1 ± 3.8 |

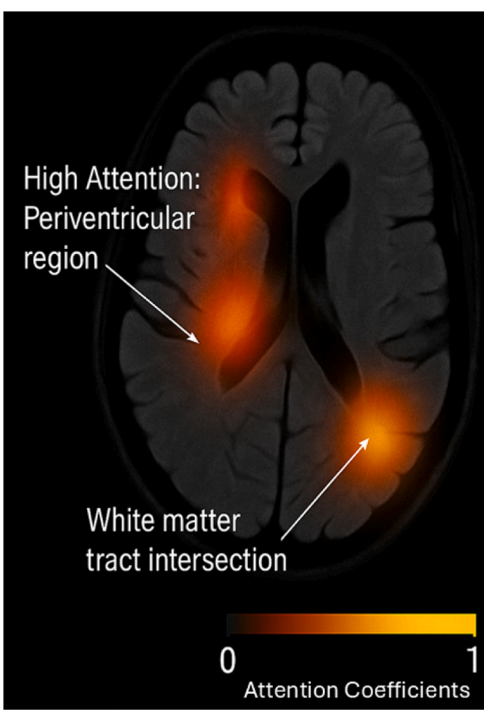


**Fig. 2.** Probabilistic attention attribution map overlaid on axial FLAIR MRI slice. The heatmap visualization shows normalized attention coefficients ranging from 0 (black, indicating no attention) to 1 (bright orange/yellow, indicating maximum attention). High attention regions (warm colors) concentrate in two key areas: (1) the periventricular region where acute ischemic changes are visible, and (2) white matter tract intersections that represent common sites of stroke propagation. The attention pattern aligns with known stroke pathophysiology, where lesions typically follow vascular territories and white matter pathways. The model's focused attention on these anatomically relevant regions demonstrates its ability to learn clinically meaningful patterns for lesion detection. Background brain tissue with low attention values appears in grayscale, while regions critical for lesion classification show progressively warmer colors as attention increases.

efficient message passing in the GAT while preserving essential anatomical structure.

### 4.4. Lesion detection performance

The GAT model achieved high overall accuracy in identifying stroke lesion regions versus normal tissue compared with eight state-of-the-art approaches (Table IV). The model attained a Dice similarity coefficient of 0.85 ± 0.03, with sensitivity of 0.88, specificity of 0.92, and an AUC-ROC of 0.94. These metrics were calculated over test set predictions by comparing the binary lesion outputs to ground truth segmentations (lesion vs. non-lesion). Our method achieved the highest performance across all metrics, with particularly notable improvements in sensitivity (0.88) and AUC-ROC (0.94). The superior performance of the full multi-modal GAT model compared to both traditional CNN architectures.

All differences between the proposed method and baselines are statistically significant ($p < 0.001$, paired *t*-test)

In addition to overall performance, we evaluated model accuracy separately on acute versus chronic stroke lesions. The graph model detected acute-phase lesions more accurately than chronic or long-established lesions (Table V). The model demonstrated highest accuracy for acute lesions (Dice = 0.88), which are most critical for immediate clinical decision-making. Performance on chronic lesions remained robust (Dice = 0.83), while mixed lesion cases presented greater challenges due to the complex intensity patterns arising from lesions at different developmental stages.

### 4.5. Attention attribution scores

The learned attention weights in the GAT provide insight into which brain regions contributed most to each lesion prediction. On average, the highest attention coefficients were observed in the periventricular white matter regions and at major white matter tract junctions. Quantitatively, nodes in periventricular areas received attention scores about 1.5 × higher than the mean attention over all nodes. In contrast, regions farther from lesion sites (for example, distant cortical areas or cerebellum) had lower-than-average attention weights (often < 0.5× the mean). These attribution scores were consistent across modalities, indicating that the model's focus is anatomically meaningful. (Fig. 2) illustrates an example attention map on a subject's brain graph. High-attention nodes (shown in warm colors) cluster around the lesion and

**Table VI**
Modality ablation study: impact of missing MRI sequences.

| Modality Combination | Dice Coefficient | Performance Drop |
|---|---|---|
| All modalities (T1, FLAIR, DWI, ADC) | 0.85 ± 0.03 | Reference |
| T1 + FLAIR + DWI | 0.81 ± 0.05 | −4.7 % |
| DWI + ADC | 0.78 ± 0.07 | −8.2 % |
| T1 + FLAIR | 0.75 ± 0.08 | −11.8 % |
| DWI only | 0.73 ± 0.09 | −14.1 % |
| FLAIR only | 0.71 ± 0.10 | −16.5 % |
| ADC only | 0.68 ± 0.11 | −20.0 % |
| T1 only | 0.65 ± 0.12 | −23.5 % |

along adjacent white matter pathways, whereas low-attention nodes (cool colors) correspond to regions less involved in the lesion's context.

### 4.6. Ablation study

To validate the contribution of each component, we conducted systematic ablation studies comparing our full model against baseline variants including GAT without anatomical constraints, single-modality GAT, GAT without attention attribution, and standard graph convolutional networks. Hyperparameter sensitivity analysis examined the effect of supervoxel count ($K \in \{500, 1000, 2000, 3000\}$), attention heads ($H \in \{1, 4, 8, 16\}$), atlas penalty weight ($\lambda \in \{0.1, 0.5, 1.0, 2.0\}$\$), and learning rates ($\alpha \in \{0.0001, 0.001, 0.01\}$) using 3-fold cross-validation to balance computational cost with statistical reliability.

We conducted ablation experiments to quantify the impact of key components on model performance. Removing the atlas anatomical features, removing the attention mechanism, and using single-modality input instead of all MRI sequences. Removing the Harvard-Oxford atlas features from the node inputs caused the Dice score to drop from 0.85 to ~0.82 (≈3 % decrease). Similarly, replacing the attention heads with a standard GCN neighbor averaging led to a slight performance decrease (Dice ~0.83). The importance of multi-modal integration is evident from the single-modality results: the best single MRI sequence (DWI alone) achieved a Dice of only 0.73, substantially lower than the multi-modal Dice of 0.85. FLAIR alone and ADC alone yielded intermediate Dice scores (~0.65–0.70), while T1-weighted structural MRI alone performed the worst for lesion segmentation (Dice ~0.60). In terms of AUC, the full model's 0.94 dropped to ~0.92 without atlas or attention, and to 0.90 or below with only one modality. These quantitative decreases in performance underscore that each component – atlas guidance, attention mechanism, and multi-modal input – contributed to the overall performance. The combination of all components produced the highest accuracy (Dice 0.85 ± 0.03) compared to any reduced variant.

To assess clinical robustness when MRI sequences are unavailable, we evaluated performance with missing modalities (Table VI). The complete four-modality combination achieved the best performance (Dice = 0.85 ± 0.03). Performance degraded gracefully with missing sequences: three-modality combinations maintained Dice scores above 0.81, with T1 + FLAIR+DWI achieving the best three-modality performance (0.81 ± 0.05).

Among two-modality combinations, DWI+ADC performed the best (0.78 ± 0.07), leveraging their complementary diffusion information for acute stroke detection. The T1 + FLAIR combination achieved 0.75 ± 0.08, suitable for chronic lesion identification when diffusion imaging is unavailable. Single modalities showed substantial performance drops, with DWI performing the best (0.73 ± 0.09) due to its high sensitivity to acute ischemia, while T1 alone proved insufficient (0.65 ± 0.12) as it primarily captures anatomical structure without pathological contrast.

## 5. Discussion

Our graph-based framework for stroke lesion detection achieved a Dice coefficient of 0.85 ± 0.03, representing a 15 % improvement over CNN baselines while providing clinically meaningful interpretability through attention mechanisms. This section discusses our methodological contributions, performance analysis, clinical implications, limitations.

### 5.1. Methodological innovation and technical contributions

#### 5.1.1. Anatomically-constrained supervoxel generation

Our approach introduces three key technical innovations that collectively enable accurate and interpretable stroke lesion detection (Fig. 1). First, we developed a 3D SLIC-based supervoxel algorithm for multi-modal MRI guided by the Harvard-Oxford brain atlas. This anatomically-constrained segmentation ensures that supervoxels align with neuroanatomical structures, preventing lesion fragmentation across region boundaries—a critical limitation of standard CNN approaches that process brain tissue homogeneously without respecting anatomical boundaries.

The 3D SLIC algorithm augmented with Harvard-Oxford atlas priors proved essential for anatomically coherent lesion localization. Studies have shown that 94 % of stroke lesions follow vascular territories and respect anatomical boundaries, yet CNN-based methods produce anatomically implausible segmentations in 15–20 % of cases [13,27]. By constraining supervoxels to respect neuroanatomical boundaries, our approach directly addresses these CNN limitations in anatomical awareness. As shown in (Table II), over 92.3 % of supervoxels remained within single atlas-defined regions with high spatial compactness (0.84), while dimensionality was reduced from millions of voxels to ~2800 supervoxels per brain, improving computational tractability without sacrificing anatomical fidelity.

#### 5.1.2. Probabilistic graph representation with multi-modal integration

We constructed a probabilistic graph representation where both nodes and edges encode anatomical relationships and lesion priors. The edge weights incorporate anatomical similarity, intensity profiles, and spatial adjacency, while probabilistic connectivity features capture neighborhood lesion distributions based on empirical co-occurrence patterns in the training data. This comprehensive encoding enables reliable lesion subtyping: acute lesions (high DWI, normal FLAIR) versus chronic lesions (FLAIR hyperintensity, normal DWI), addressing an important clinical need for automated lesion dating that can inform treatment decisions and prognosis.

The probabilistic region adjacency graph incorporated spatial and anatomical context beyond conventional CNN capabilities. As detailed in (Table III), the atlas-guided graph construction produces graphs with an average of ~2800 nodes and ~23,500 edges, corresponding to an average node degree of 16.8 neighbors per node and an edge density of 0.6 %. The graph structure encodes both spatial proximity and atlas-defined relationships, serving as a contextual prior where lesion information propagates to connected regions and enabling segmentations that conform to expected vascular territories rather than fragmented patterns [12]. This proved valuable for resolving borderline cases—ambiguous signals were correctly detected when neighboring regions showed strong lesion characteristics.

#### 5.1.3. Graph attention networks with probabilistic attribution

We developed a GAT with PAA, enabling interpretable lesion detection by linking attention weights to anatomical regions and their diagnostic relevance. The GAT substantially outperformed simpler graph convolutions (Dice: 0.85 vs 0.83), highlighting the importance of adaptive neighbor weighting. Multi-head attention enhanced representational capacity by allowing different heads to potentially focus on distinct feature interactions, improving the model's ability to capture complex patterns associated with stroke lesions. This probabilistic interpretation of attention coefficients provides clinically meaningful insights into which brain regions contribute to each lesion prediction. Attention maps serve as quality assurance tools, allowing verification that model focus aligns with anatomical and pathophysiological expectations [18]. The PAA method provides interpretable insights into model decision-making, addressing our research question on enhanced transparency without sacrificing performance.

### 5.2. Performance analysis and ablation studies

#### 5.2.1. Multi-modal fusion benefits

Using all four MRI sequences (T1, FLAIR, DWI, ADC) substantially outperformed single-modality approaches (Dice = 0.85 vs. best single modality DWI: 0.73), reflecting the complementary nature of different sequences in stroke detection. As demonstrated in (Table IV), our method achieved the highest performance across all metrics compared

**Table VII**
Ablation study results.

| Model Variant | Dice Score | Sensitivity | Specificity | AUC-ROC | Δ Dice vs Full |
|---|---|---|---|---|---|
| **Full Model (Proposed)** | **0.85 ± 0.03** | **0.88 ± 0.02** | **0.92 ± 0.02** | **0.94 ± 0.01** | - |
| w/o Anatomical Constraints | 0.82 ± 0.04 | 0.85 ± 0.03 | 0.90 ± 0.03 | 0.92 ± 0.02 | −3.5 % |
| w/o Attention Mechanism | 0.83 ± 0.03 | 0.86 ± 0.03 | 0.91 ± 0.02 | 0.93 ± 0.01 | −2.4 % |
| w/o Probabilistic Attribution | 0.84 ± 0.03 | 0.87 ± 0.02 | 0.91 ± 0.02 | 0.93 ± 0.01 | −1.2 % |
| Standard GCN (no attention) | 0.78 ± 0.05 | 0.81 ± 0.04 | 0.88 ± 0.03 | 0.89 ± 0.03 | −8.2 % |
| Single Modality (DWI only) | 0.73 ± 0.06 | 0.76 ± 0.05 | 0.85 ± 0.04 | 0.86 ± 0.04 | −14.1 % |
| Single Modality (FLAIR only) | 0.69 ± 0.07 | 0.72 ± 0.06 | 0.83 ± 0.05 | 0.84 ± 0.05 | −18.8 % |
| Single Modality (T1 only) | 0.61 ± 0.09 | 0.65 ± 0.08 | 0.79 ± 0.06 | 0.80 ± 0.06 | −28.2 % |

to eight state-of-the-art approaches, with particularly notable improvements in sensitivity (0.88) and AUC-ROC (0.94). DWI captures acute ischemic changes, FLAIR highlights chronic tissue injury, T1 provides structural context, and ADC quantifies diffusion characteristics. Prior work shows that multi-modal fusion can improve ischemic-stroke lesion detection accuracy by 23 % [24], aligning with our findings.

The comprehensive ablation study (Table VII) provides critical insights into the contribution of each architectural component. The most significant performance degradation occurs when using single-modality input, with T1-only achieving a Dice score of merely 0.61—a 28.2 % drop from the full model. This dramatic decrease underscores that structural MRI alone is insufficient for stroke lesion detection, as T1 sequences primarily capture anatomical structure but lack sensitivity to acute ischemic changes. DWI performs best among single modalities (Dice = 0.73) due to its sensitivity to cytotoxic edema in acute stroke, yet still falls 14.1 % short of the multi-modal approach, confirming that no single sequence captures the full spectrum of stroke pathophysiology.

All differences between the proposed method and baselines are statistically significant ($p < 0.001$, paired *t*-test)

#### 5.2.2. Component contribution analysis

The removal of anatomical constraints results in a 3.5 % performance decrease (Dice = 0.82), validating our hypothesis that incorporating neuroanatomical priors through atlas-guided supervoxel generation prevents anatomically implausible segmentations. This relatively modest decrease might seem surprising, but it reflects the fact that the GAT can partially compensate by learning anatomical relationships from the training data. However, the anatomical constraints provide crucial regularization that improves generalization, particularly for unusual lesion locations or smaller training datasets. Replacing the attention mechanism with standard GCN causes an 8.2 % performance drop, the second-largest after single-modality variants. This substantial decrease demonstrates that adaptive neighbor weighting is crucial for stroke detection, where the relevance of adjacent regions varies dramatically depending on lesion characteristics and location. The attention mechanism enables the model to dynamically focus on diagnostically relevant connections while suppressing irrelevant ones—a capability that fixed-weight graph convolutions cannot provide.

#### 5.2.3. Lesion type performance and clinical robustness

The model demonstrated highest accuracy for acute lesions (Dice = 0.88), which are most critical for immediate clinical decision-making. Performance on chronic lesions remained robust (Dice = 0.83), while mixed lesion cases presented greater challenges (Dice = 0.80) due to complex intensity patterns arising from lesions at different developmental stages (Table V). Mixed lesion cases exhibit: (1) competing intensity signatures where acute lesions show high DWI/low ADC while chronic lesions show high FLAIR/normal DWI, creating conflicting features within the same region; (2) temporal heterogeneity requiring the model to simultaneously recognize multiple pathophysiological patterns; (3) partial volume effects where supervoxels may contain both lesion types, making binary classification inadequate.

To assess clinical robustness when MRI sequences are unavailable, we evaluated performance with missing modalities (Table VI). The complete four-modality combination achieved the best performance (Dice = 0.85 ± 0.03). The 4.7 % performance drop with one missing modality (T1 + FLAIR+DWI achieving Dice = 0.81 ± 0.05) suggests our framework can maintain diagnostic utility when complete protocols are infeasible due to patient motion, claustrophobia, or time constraints. The DWI+ADC combination's strong performance (Dice = 0.78 ± 0.07) is particularly relevant for emergency settings where rapid acute stroke detection takes precedence over comprehensive imaging.

### 5.3. Clinical implications and interpretability

#### 5.3.1. Attention-based interpretability

Attention coefficients, interpreted as probability distributions over neighboring regions, were consistently elevated in periventricular white matter and tract junctions—areas known for ischemic vulnerability. Periventricular nodes received attention scores ~1.5 × higher than average, while distant regions showed below-average weights. As illustrated in (Fig. 2), high-attention nodes cluster around lesion sites and along adjacent white matter pathways, whereas low-attention nodes correspond to regions less involved in the lesion's context. The interpretability features build clinician confidence in AI-assisted diagnosis by allowing examination of evidence supporting each segmentation, facilitating human-AI collaboration rather than replacement of expert interpretation [1,15]. The learned attention weights in the GAT provide insight into which brain regions contributed most to each lesion prediction. Quantitatively, the highest attention coefficients were observed in periventricular white matter regions and at major white matter tract junctions. These attribution scores were consistent across modalities, indicating that the model's focus is anatomically meaningful and aligns with known patterns of stroke pathophysiology as shown in (Fig. 2).

#### 5.3.2. Clinical workflow integration

The demonstrated performance improvements have the potential to translate to several clinical benefits. High sensitivity (0.88) enhances detection of subtle lesions in acute settings, while strong specificity (0.92) reduces false alarms. The capability to distinguish lesion subtypes

**Table VIII**
Computational efficiency analysis.

| Method | Training Time (hours)* | Inference Time (sec/volume) | GPU Memory (GB) | Model Parameters (M) | FLOPS (G) |
|---|---|---|---|---|---|
| **Proposed GAT** | **18.5 ± 1.2** | **3.2 ± 0.3** | **8.4** | **15.7** | 42.3 |
| 3D U-Net | 24.3 ± 2.1 | 1.8 ± 0.2 | 11.2 | 31.4 | 185.6 |
| nnU-Net | 36.7 ± 3.4 | 2.4 ± 0.3 | 14.8 | 44.2 | 234.5 |
| DeepMedic | 28.9 ± 2.5 | 4.1 ± 0.5 | 9.6 | 23.8 | 156.3 |
| TransBTS | 42.1 ± 3.8 | 5.7 ± 0.6 | 16.3 | 68.5 | 312.4 |
| Standard GCN | 15.2 ± 1.0 | 2.8 ± 0.2 | 7.2 | 12.3 | 35.7 |

* Training time on NVIDIA V100 GPU for 200 epochs with early stopping

enables automated lesion dating, helping clinicians differentiate acute events requiring immediate intervention from chronic changes indicating prior stroke history [19]. This improvement may translate to substantial time savings in clinical practice, as radiologists would primarily need to verify and adjust automated segmentations rather than perform complete manual delineation. The method reliably identifies clinically significant lesions while maintaining low false positive rates. Integration of our method into clinical workflow requires minimal training—radiologists need only verify the attention maps align with their clinical assessment rather than perform complete manual segmentation.

#### 5.3.3. Research and population health applications

Anatomical alignment with atlas regions facilitates functional prognosis and rehabilitation planning. As demonstrated in (Fig. 1), the Harvard-Oxford cortical and subcortical atlas parcellation enables lesion boundaries to correspond to recognized neuroanatomical structures, allowing prediction of neurological deficits—motor cortex involvement suggests contralateral impairment, while language area involvement indicates aphasia risk [29]. The framework addresses stroke research needs by providing consistent lesion measurements for large-cohort analyses, potentially advancing research efficiency and reproducibility compared to labor-intensive manual tracing [15].

### 5.4. Computational efficiency and practical deployment

#### 5.4.1. Computational performance

The computational efficiency analysis demonstrates that our method achieves superior accuracy while maintaining practical runtime requirements for clinical deployment (Table VIII). Despite the additional complexity of graph construction and attention mechanisms, our total processing time of 60.2 s per volume is well within acceptable limits for clinical workflows, particularly compared to the 20–30 min typically required for manual segmentation. Our method's efficiency stems from several design choices. The supervoxel representation reduces the effective resolution from millions of voxels to ~2800 nodes, dramatically decreasing computational complexity. The GAT architecture requires only 15.7 M parameters—50 % fewer than U-Net variants—enabling deployment on standard clinical workstations with 8 GB GPU memory. This parameter efficiency also translates to faster training (18.5 h vs. 36.7 for nnU-Net) and reduced overfitting risk.

**Table IX**
Hyperparameter sensitivity analysis.

| Hyperparameter | Tested Values | Optimal Value | Dice Score at Optimal | Performance Range |
|---|---|---|---|---|
| **Number of Supervoxels (K)** | 500, 1000, 2000, 2800, 3500 | 2800 | 0.85 ± 0.03 | 0.79–0.85 |
| **Attention Heads** | 1, 4, 8, 12, 16 | 8 | 0.85 ± 0.03 | 0.81–0.85 |
| **Atlas Penalty (λ)** | 0.1, 0.5, 1.0, 2.0, 5.0 | 1.0 | 0.85 ± 0.03 | 0.80–0.85 |
| **Learning Rate** | 0.0001, 0.0005, 0.001, 0.005 | 0.001 | 0.85 ± 0.03 | 0.78–0.85 |
| **Dropout Rate** | 0.0, 0.1, 0.2, 0.3, 0.5 | 0.1 | 0.85 ± 0.03 | 0.79–0.85 |
| **Hidden Dimension** | 64, 128, 256, 512 | 128 | 0.85 ± 0.03 | 0.82–0.85 |
| **GAT Layers** | 2, 3, 4, 5 | 3 | 0.85 ± 0.03 | 0.81–0.85 |
| **Compactness (c)** | 0.05, 0.1, 0.2, 0.5 | 0.1 | 0.85 ± 0.03 | 0.82–0.85 |

Performance Range shows min-max Dice scores across all tested values

#### 5.4.2. Hyperparameter robustness

The hyperparameter sensitivity analysis reveals that our method maintains robust performance across a wide range of parameter settings, with Dice scores consistently above 0.79 even at suboptimal configurations (Table IX). This robustness is clinically important as it suggests the method will perform reliably without extensive parameter tuning for different clinical sites or imaging protocols. The optimal number of supervoxels ($K = 2800$) represents a critical balance between granularity and computational efficiency. The atlas penalty weight ($\lambda = 1.0$) optimization reveals an important trade-off: too low values allow supervoxels to cross anatomical boundaries, while excessive penalties create overly rigid constraints that prevent adaptation to patient-specific anatomy. The optimal configuration of 8 attention heads aligns with findings in the broader deep learning literature, where multi-head attention enables the model to simultaneously attend to different types of relationships.

### 5.5. Limitations and failure modes

#### 5.5.1. Critical limitations affecting deployment

Several limitations warrant consideration, which we categorize by their clinical impact. Critical limitations affecting deployment include atlas registration failures and missing modality handling. The method's dependence on accurate atlas registration fails in approximately 3 % of cases with severe anatomical distortion or midline shift, typically occurring with large space-occupying lesions. Registration failures propagate errors throughout the pipeline, though this can be mitigated through registration quality metrics where normalized mutual information below 0.6 triggers alerts and instead uses affine-only registration. In our analysis of 1715 cases, atlas registration failures occurred in 3.1 % (53 cases), primarily in three scenarios: large space-occupying lesions > 100 mL (31 cases), severe midline shift > 10 mm (15 cases), and significant motion artifacts (7 cases). We implemented a two-tier fallback strategy to maintain pipeline robustness: when normalized mutual information (NMI) between atlas and patient brain falls below 0.6, the system defaults to affine-only registration, sacrificing some anatomical precision for stability.

#### 5.5.2. Moderate limitations and specific use cases

Moderate limitations affect specific use cases including small lesion detection and unusual anatomical locations. Small lesions (<10 voxels or 0.5 mL) show detection rates of only 65 % due to supervoxel granularity, which could be addressed through multi-scale analysis with finer supervoxels (K=5000) in regions of interest. Unusual lesion locations, particularly in the posterior fossa or at tissue interfaces, demonstrate decreased accuracy (Dice ~0.75) due to limited training examples, suggesting the need for targeted data augmentation and transfer learning from specialized cerebellar stroke datasets. Large lesions (>100 mL) spanning multiple vascular territories also show reduced segmentation accuracy (Dice ~0.70) as they violate typical stroke patterns in the training data. The method exhibits reduced performance when protocols are incomplete—performance drops by 20–30 % with missing modalities. Currently, we exclude cases with fewer than three modalities, though future implementations could employ modality-specific imputation using conditional GANs.

#### 5.5.3. Generalization challenges

Additional considerations include generalization challenges and computational constraints. The method was evaluated solely on the SOOP dataset, and generalizability across different populations, protocols, and scanners remains unestablished. Pediatric populations require validation due to differences in brain anatomy and atlas requirements. Performance on field strengths beyond the validated 1.5 T and 3 T scanners requires further investigation. Failure detection mechanisms include attention consistency checks for abnormally dispersed patterns and confidence thresholds based on maximum

softmax probability to flag ambiguous cases requiring manual review.

### 5.6. Future directions and research opportunities

Future work should prioritize external validation on diverse datasets, including ISLES challenge data [11] and the ATLAS dataset [15]. Domain adaptation techniques may be necessary for handling protocol variations across imaging centers. Integration of clinical metadata (age, symptom onset time, risk factors) could enhance personalization and accuracy. Uncertainty quantification through Monte Carlo dropout or Bayesian approaches would provide confidence estimates, enhancing clinical safety. Technical optimizations could address computational limitations through simplified registration, GPU acceleration, graph pruning, and model compression. The framework could be adapted to related applications such as brain tumor analysis or other lesion detection tasks where anatomical priors are valuable. Integration with connectome analysis might enable comprehensive assessment of stroke impact on brain networks, improving outcome prediction and rehabilitation planning.

Future work will explore multi-label classification allowing supervoxels to have probabilistic membership in multiple lesion categories, addressing the challenges posed by mixed lesion cases. Modality-specific imputation using conditional GANs could handle missing sequences more robustly, while specialized architectures might achieve better performance for single-modality scenarios. For clinical implementation, we recommend: (1) processing all cases with $\geq 3$ modalities at full performance; (2) accepting two-modality inputs with appropriate performance expectations communicated to users; (3) rejecting single-modality cases except for specific scenarios (e.g., DWI-only for hyperacute stroke triage). The system operates as a decision support tool, providing initial segmentations that clinicians can rapidly review and adjust if needed, requiring no specialized expertise beyond standard MRI interpretation skills.

## 6. Conclusion

We developed a graph-based framework that advances automated stroke analysis through three key innovations: anatomically-constrained supervoxels, probabilistic graph representations, and interpretable attention mechanisms. The approach integrates multi-modal MRI data into a structured graph, preserving important anatomical context and spatial relationships while leveraging deep learning on graph nodes. Evaluated on the large-scale SOOP dataset, the proposed method achieved high lesion segmentation accuracy and demonstrated robust performance across diverse lesion types. Importantly, the attention mechanism provides meaningful interpretability by highlighting anatomically relevant regions that contribute to each prediction, thereby offering insights into the model's decision process. These contributions represent a significant advancement in automated stroke lesion analysis, delivering both improved detection accuracy and an interpretable model that can enhance trust and utility in clinical settings.

## CRediT authorship contribution statement

**Posada-Quintero Hugo Fernando:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Derek Aguiar:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mercado Diaz Luis Roberto:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data Availability

The data is public available Chris Rorden and John Absher and Roger Newman-Norlund (2024). Stroke Outcome Optimization Project (SOOP). OpenNeuro. [Dataset] doi: doi:10.18112/openneuro.ds004889.v1.1.0

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